6.11 Thermal characteristics

The operating junction temperature T_J must never exceed the maximum given in *Table 24: General operating conditions*.

The maximum junction temperature in °C that the device can reach if respecting the operating conditions, is:

$$T_J(max) = T_A(max) + P_D(max) \times \Theta_{JA}$$

where:

- $T_A(max)$ is the maximum operating ambient temperature in °C,
- Θ_{JA} is the package junction-to-ambient thermal resistance, in °C/W,
- $P_D = P_{INT} + P_{I/O}$,
  - P_{INT} is power dissipation contribution from product of I_{DD} and V_{DD}
  - $P_{I/O}$ is power dissipation contribution from output ports where $P_{I/O} = \Sigma (V_{OL} \times I_{OL}) + \Sigma ((V_{DDIOx} - V_{OH}) \times I_{OH})$, taking into account the actual V_{OL} / I_{OL} and V_{OH} / I_{OH} of the I/Os at low and high level in the application.

Table 81. Thermal resistance

Symbol	Parameter	Package(1)	Value	Unit
Θ_{JA}	Thermal resistance junction-ambient	WLCSP19	101.9	°C/W
		TSSOP20	81.4	
		UFQFPN28	59	
		UFQFPN32	42.9	
		LQFP32	54.3	
		UFQFPN48	32.4	
		LQFP48	54.3	
		LQFP64	47.2	
		UFBGA64	69.1	
Θ_{JB}	Thermal resistance junction-board	WLCSP19	74.6	°C/W
		TSSOP20	52.2	
		UFQFPN28	23.8	
		UFQFPN32	24.8	
		LQFP32	31.6	
		UFQFPN48	16.7	
		LQFP48	31.6	
		LQFP64	29.6	
		UFBGA64	52	